

MIDTERM EXAMINATION **ANSWERS**

ECON 201, Spring 2014
Intermediate Macroeconomics
Prof. Ryan Edwards

March 10, 2012

RULES:

- Don't start until I say so.
- You may check that you have all 11 pages.
- Write your name and **seat number** at the top *and on page 7*.
- When I call time, **stop**. If you don't, you get an F.
- No books, no notes, no calculators, **no electronic devices**.
- This exam has 1 whimsical warmup and 4 main questions together worth 75 points, one point for each minute of time you have. Budget your time.
- If I catch you cheating in any way, you will receive an F for the course and a referral to the VP for Student Affairs.

BE GOOD, NEVER BE EVIL

Points:	1	5	2	3	4	5	Total:
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1. (5 points total) *Whimsical Warmup.*

THERE ARE NO WRONG ANSWERS. You will get all 5 points even if you don't answer anything. See the large "5" above?

- (a) (1 point) Bruno Mars performed the Super Bowl Halftime Show this year. What did you think?
- i. Sally Venus would have been better
 - ii. Mmmmm, Mars bar
 - iii. Sweet
 - iv. Kind of dull but also intergenerationally appealing. I just really like that nice young man
- (b) (1 point) The Denver Broncos:
- i. Splat
 - ii. Boom!
 - iii. Who?
 - iv. Yes.
- (c) (1 point) Miley Cyrus and Justin Bieber:
- i. Weigh about 100 lbs *combined*
 - ii. Don't tweet enough, really
 - iii. One is jealous of the other's last name. "Sigh" just isn't a cool verb like "Believe"
 - iv. Two great things that would go great together, but oddly they haven't
- (d) (1 point) The food on campus at Queens College
- i. Is super yummy
 - ii. There's food on campus?
 - iii. Is so amazing *because* of all the off-campus competition from great local restaurants
 - iv. Technically is all off-campus in food trucks. Thankfully
- (e) (1 point) Econ 201 is
- i. Burp
 - ii. My BFF of econ courses. I heart you, Econ 201
 - iii. Something that keeps me awake at night
 - iv. Thankfully half over

2. (10 points total) *Short Answer Potpourri.*

- (a) (2 points) Write a sentence describing what “capital” is in Econ 201.

Answer: Capital is physical stuff that produces other stuff. If you said “stocks,” “bonds,” or “money,” that’s half right; those can represent claims on capital. The key concept for Econ 201 is that capital — machinery and other physical goods — combines with labor to produce output.

- (b) (2 points) Write a sentence describing what this equation means in Econ 201, using words like “production” or “income.”

$$Y = AK^{1/3}L^{2/3}$$

Answer: A country’s production, output, or income (Y) is produced by the combination of technology or productivity (A), capital (K), and labor (L).

- (c) (2 points) Recall that nominal GDP (
- Y^N
-) is related to real GDP (
- Y^R
-) and the price level (
- P
-) in the following way:

$$Y^N = P Y^R$$

If $g[P] = 0.03$ and $g[Y^R] = 0.02$, what is $g[Y^N]$?

Answer: Taking growth rates of both sides, we find that $g[Y^N] = g[P] + g[Y^R]$. Thus if $g[P] = 0.03$ and $g[Y^R] = 0.02$, then it must be true that $g[Y^N] = 0.03 + 0.02 = 0.05$.

- (d) (2 points) If I deposited \$100 into a bank account that paid 2% interest annually, how much would I have accumulated after 2 years? You do not need a calculator, just write down a mathematical expression.

Answer: $100(1 + 0.02)^2$

- (e) (2 points) The marginal product of labor is given by

$$MPL = \frac{2}{3}A \left(\frac{K}{L} \right)^{1/3}$$

If technology (A) increases by 5%, what happens to the wage, if anything? Be specific.

Answer: The short answer, for half credit, is that the wage rises when A increases. The long answer is obtained by taking growth rates of both sides: $g[MPL] = g\left[\frac{2}{3}\right] + g[A] + g\left[\left(\frac{K}{L}\right)^{1/3}\right]$. If only A is increasing, then the other growth rates are zero, and we are left with $g[MPL] = g[A] = 0.05$.

3. (25 points total) *Famine and out-migration in the Solow Model.*

St. Patrick's Day 2014 is next Monday. In 2011, 34.5 million U.S. residents claimed Irish ancestry, meaning there were 7 Irish Americans for every single resident of Ireland today, which had a population of 4.68 million.¹ What events could have led to such a situation?

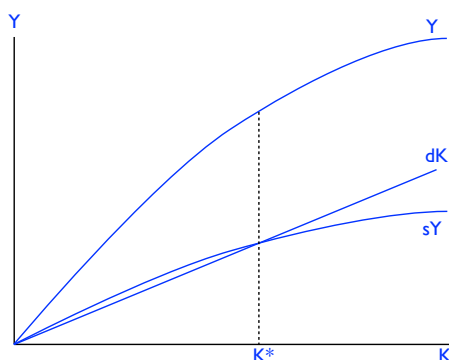
The Great Famine of 1846–47 in Ireland was caused by a massive failure of the potato crop due to infestation by a *potato blight*, a disease that kills potatoes. How can we view this event and its implications from the perspective of the Solow Model?

Recall that the key equations of the Solow Model are:

$$Y = A K^{1/3} L^{2/3} \quad (1)$$

$$\Delta K = sY - dK \quad (2)$$

- (a) (4 points) On the axes below, draw the Solow diagram for 1845 Ireland, identifying the saving curve, depreciation curve, income curve, and the steady state.



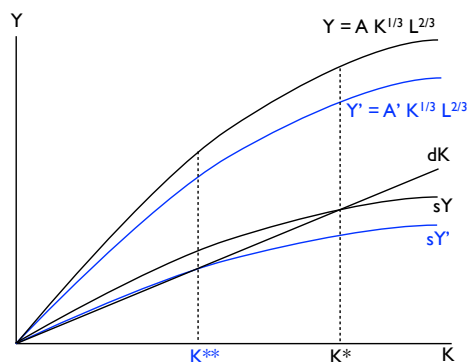
Unlike a plague or virus that kills people, the potato blight kills potatoes. Even with the same amount of capital available, potato farmers were vastly less productive after the blight arrived because the fruits of their labor simply decayed away.

- (b) (3 points) What parameter in the Solow Model can we use to model the effects of the potato blight? What does the blight do to that parameter? Explain.

Answer: It's the productivity parameter A that we can use to show the effects of a potato blight that vastly reduce the productivity of farmers (L) given the amount of capital (K) they have. A potato blight thus reduces A dramatically.

¹http://www.census.gov/newsroom/releases/pdf/cb13-ff03_st_patricks_day.pdf

- (c) (4 points) Redraw the Solow diagram depicting the change you described in part (b) for 1846 Ireland, identifying the saving curve(s), depreciation curve(s), income curve(s), and the steady state(s). If they are different, show “before” and “after” versions as appropriate, and briefly explain what you have drawn.



Answer: A drop in productivity A lowers both the income and saving curves and produces a new lower steady state at K^{**} .

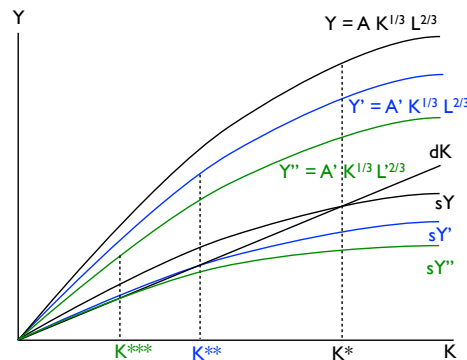
- (d) (3 points) What will happen in the short run to income per person (Y/L) as a consequence of what you have shown in part (c)? What about in the long run? Explain.

Answer: Income per person immediately falls because of the reduction in A , and it will fall even further in the long run because of the reduction in steady-state K from K^* to K^{**} .

- (e) (3 points) How would you advise an Irish resident who was facing what you described in part (d)? Would you suggest that he or she migrate to the United States, or stay put in Ireland? Remember that crossing the Atlantic Ocean was very costly in terms of time and money in 1846. Explain fully.

Answer: It may be costly to migrate out of Ireland, but staying put is an even more costly prospect because income *every year* is vastly reduced. Paying a fixed one-time cost to migrate sounds like a relatively good idea, even if the cost is relatively high. As long as the fixed cost can be recouped over future periods of life, the better decision would be to migrate.

- (f) (4 points) Supposing that a large number of Irish residents took the advice you gave in part (e), illustrate the joint effects of that change along with the original change you showed in part (c), and explain. Discuss the net effect on income (Y).



Answer: If a bunch of Irish workers migrated out, that would reduce L in the Solow Model. A reduction in L looks exactly the same as a reduction in A : they both shift the income and saving curves down simultaneously. The net effect on income is a BIG reduction.

In the Solow Model, income per person in the short run equals

$$\frac{Y}{L} = A \left(\frac{K}{L} \right)^{1/3}$$

while in the long run, in the steady state, it equals

$$\frac{Y^*}{L} = A^{1/2} \left(\frac{s}{d} \right)^{1/2}$$

- (g) (4 points) Discuss the short-run and long-run effects of the development you depicted above *in part (f)* on income per Irish person (Y/L). What are they? Are they the same, or are they different? Explain and discuss using some intuition.

Answer: While out-migration (that's the development from part (f)) decreases GDP, it actually *raises* GDP per person in the short run, because capital is fixed while labor falls. Meanwhile, in the long run, out-migration has *no effect* on GDP per person. It doesn't affect Y^*/L at all. The intuition for these interesting effects are that when people leave, labor becomes more scarce, and output per worker, or average productivity, rises because there is more capital per worker. But in the long run, capital adjusts downward because there are fewer people around.

4. (10 points) *A Model Immigrant*

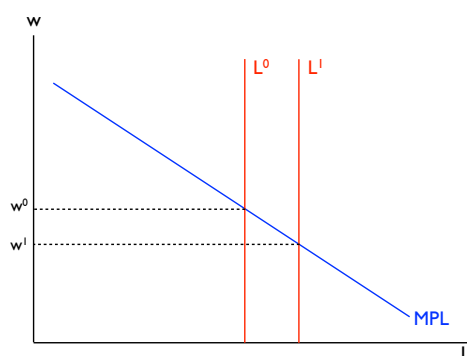
Recall that in the simple neoclassical production model of chapter 4, the marginal product of labor is given by:

$$MPL = \frac{2}{3}A \left(\frac{K}{L} \right)^{1/3} \quad (3)$$

- (a) (2 points) Qualitatively, what is the first-order effect of the arrival of more immigrants who arrive to work? That is, what does immigration change, if anything, in the simple model of production?

Answer: The first thing immigration changes is the labor supply, L , which must increase.

- (b) (4 points) On the axes below, draw the labor demand curve(s), vertical labor supply curve(s), and show the market-clearing wage(s) in the labor market both before and after a number of immigrants arrive.



- (c) (2 points) If immigrants bring capital with them or save and produce capital after they arrive, how might that change the story, if at all? Explain, with references to the graph.

Answer: If either of those things were to happen, then the capital stock would respond to immigration as well. Capital raises the marginal productivity of labor across the board, shifting the labor demand curve out to the right.

- (d) (2 points) If immigrants save after they arrive and send their savings back to family residing in their countries of origin, how might that change the story, if at all? Explain, with references to the graph.

Answer: If immigrants sent their saving overseas, it would not become domestic capital unless whoever received the transfer took it and reinvested it in U.S. companies. The latter is possible, but if it doesn't happen, then immigrants' arrival does not produce more domestic capital, and the graph we drew in part (b) is pretty much the story.

Points:	1	5	2	3	4	5	Total:
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5. (25 points) *Government, R&D, and the Romer Model.*

Does a Romer economy occur naturally, through market forces? Put another way, if all factors are paid their marginal products, will sustainable growth occur naturally?

Recall that the Romer model includes these equations:

$$\Delta A_t = \bar{z} A_t L_{at} \quad (4)$$

$$Y_t = A_t L_{yt} \quad (5)$$

$$\bar{N} = L_{at} + L_{yt} \quad (6)$$

$$L_{at} = \bar{\ell} \bar{N} \quad (7)$$

For simplicity, let's call L_{yt} "farmers" and L_{at} "scientists."

Part I: Solving the standard Romer Model.

(On this page, you may "skip to the end" and not follow the steps as I have laid them out, **if** you show your work.)

- (a) (2 points) Use equations (4) and (7) to find an equation for the growth rate of A in terms of $\bar{z}$, $\bar{\ell}$, and $\bar{N}$.

Answer: In whichever order you like, divide both sides of equation (4) by A_t to find a formula for the growth rate of A , g_A , and use equation (7) to substitute for L_{at} :

$$\begin{aligned} \frac{\Delta A_t}{A_t} &= \frac{\bar{z} A_t L_{at}}{A_t} \\ g_A &= \bar{z} L_{at} \\ &= \bar{z} \bar{\ell} \bar{N} \end{aligned}$$

- (b) (2 points) Use equations (5), (6), and (7) to find an equation for income per person, $Y/\bar{N}$, in terms of A_t and $\bar{\ell}$.

Answer: Divide both sides of equation (5) by $\bar{N}$, and substitute for L_{yt} using equations (6) and (7), which show that $L_{yt} = (1 - \bar{\ell})\bar{N}$:

$$\begin{aligned} \frac{Y}{\bar{N}} &= \frac{A_t L_{yt}}{\bar{N}} \\ &= \frac{A_t (1 - \bar{\ell}) \bar{N}}{\bar{N}} \\ &= A_t (1 - \bar{\ell}) \end{aligned}$$

- (c) (2 points) Use your answer to part (a) to find an equation for A_t using its starting level A_0 , and then substitute for A_t in your answer to part (b) in order to find an equation for income per person, $Y/\bar{N}$, solely in terms of the parameters: A_0 , $\bar{z}$, $\bar{\ell}$, and $\bar{N}$.

Answer: If A_t is growing at a constant rate g_A , then for some starting level A_0 , we know that $A_t = A_0(1 + g_A)^t$. Substituting that and our expression for g_A into the previous answer gets us:

$$\begin{aligned}\frac{Y}{\bar{N}} &= A_t(1 - \bar{\ell}) \\ &= A_0(1 + \bar{z}\bar{\ell}\bar{N})^t(1 - \bar{\ell})\end{aligned}\tag{8}$$

Part II: Interpreting the Romer Model.

The marginal product of farmers, L_{yt} , is equal to A_t . That is also their average product because there are constant returns to labor in goods production in this standard version of the Romer Model.

- (d) (2 points) If farmers, L_{yt} , are paid their marginal product as their wage, w_y , write down a formula for the total wages paid to labor in goods production.

Answer: We are told that farmers' marginal product is A_t . So the total wages paid to farmers must be that times L_{yt} , or $w_t L_{yt} = A_t L_{yt}$.

- (e) (2 points) Subtract your answer to part (d) from total goods production given by equation (5), which should equal the total amount of output that is paid to other factors of production. Briefly explain what you have found.

Answer: This may confuse you, but here's what happening: total goods production $Y_t = A_t L_{yt}$. But at the same time, that is also exactly equal to the answer to part (d), the total wages paid to labor. So the difference, which equals the total amount paid to other factors of production, is $Y_t - w_t L_{yt} = A_t L_{yt} - A_t L_{yt} = 0$. Zero?!?! Yes, true story! Read on.

In addition to farmers, L_{yt} , there are scientists, L_{at} . The marginal product of scientists, in terms of new ideas, is equal to $\bar{z}A_t$. That is, competitive markets would pay scientists in ideas.

- (f) (2 points) Can scientists eat their ideas? What would they have to do with their ideas in order to survive? Explain.

Answer: They can't eat their ideas, silly grasshopper. Scientists would have to trade their ideas for food to eat in order to survive, because they are not producing any food.

- (g) (2 points) Scientists are paid in ideas, but do they really "own" those ideas they are paid? Can they trade them for other things? Explain.

Answer: They don't really own their marginal product, because ideas are nonrival and can't really be (easily) owned or traded. Ideas can be costlessly communicated and reproduced, and thus they aren't a good store of value at all.

- (h) (3 points) Think about what you have found. If competitive markets determine the allocation of labor, to what level would the scientist share $\bar{\ell}$ quickly converge? Why?

Answer: Zero. Scientists paid their marginal product would have no earnings in terms of food. They'd die, or be smart enough not to become scientists in the first place!

- (i) (3 points) From the perspective of a “newborn” individual deciding to become either a farmer or a scientist, what would have to be true about the levels of the wages in those two fields to make that individual stick with his or her occupational choice rather than switch?

Answer: The farmer wage would have to be no higher than the scientist wage, and vice-versa, for individuals not to switch to the higher-earning industry. This is a standard long-run equilibrium or “no-arbitrage” argument.

Imagine government levied a tax rate τ on farmers, so that their after-tax wage was $w'_y = (1 - \tau)A_t$ and that total tax collections were $\tau A_t L_{yt}$. Suppose the government divided these collections equally among the L_{at} scientists, so that scientists earned a wage equal to

$$w'_a = \frac{\tau A_t L_{yt}}{L_{at}} \quad (9)$$

- (j) (5 points) What tax rate τ would support a scientist share of $\bar{\ell}$? [Hint: set $w'_a = w'_y$ and solve, using equation (6) to substitute for L_{yt} and L_{at} in terms of $\bar{\ell}$ and $\bar{N}$.]

Answer: The answer is that choosing $\tau = \bar{\ell}$ equalizes the wages. Why? Here is the math, starting with the assumption that the wages are equal and working from there:

$$\begin{aligned} w'_a &= w'_y \\ \frac{\tau A_t L_{yt}}{L_{at}} &= (1 - \tau)A_t \end{aligned}$$

Cancel the common A_t terms on both sides, and substitute for $L_{yt} = (1 - \bar{\ell})\bar{N}$ and $L_{at} = \bar{\ell}\bar{N}$, which produces

$$\frac{\tau(1 - \bar{\ell})\bar{N}}{\bar{\ell}\bar{N}} = (1 - \tau)$$

Cancel the $\bar{N}$ terms in the numerator and denominator, and that gives you

$$\begin{aligned} \frac{\tau(1 - \bar{\ell})}{\bar{\ell}} &= (1 - \tau) \\ \frac{1 - \bar{\ell}}{\bar{\ell}} &= \frac{1 - \tau}{\tau} \end{aligned}$$

It should be clear now that $\tau = \bar{\ell}$, but to see it, you could also simplify to

$$\begin{aligned}\frac{1}{\bar{\ell}} - 1 &= \frac{1}{\tau} - 1 \\ \frac{1}{\bar{\ell}} &= \frac{1}{\tau} \\ \bar{\ell} &= \tau\end{aligned}$$